

# ***The origin of ferromagnetism and the Hubbard model***

## ***part 4: Nagaoka ferromagnetism***

***Advanced Topics in  
Statistical Physics  
by Hal Tasaki***

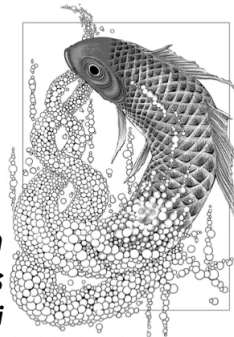


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Nagaoka 1966

somewhat singular but first rigorous example of ferromagnetism in the Hubbard model with

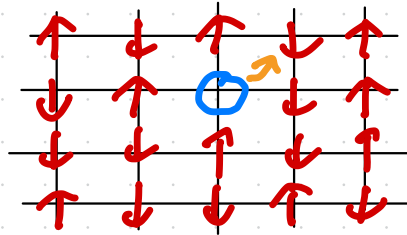
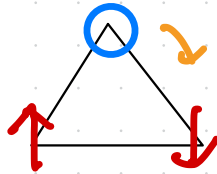
$U = \infty$  and  $N = |\Lambda| - 1$

no double occupancy

exactly one "hole"

(Stoner criterion  $UD_F > 1$ )

example in part 2



motion of a single "hole" generates global ferromagnetism

$$\hat{H} = \sum_{\substack{x, y \in \Lambda \\ \sigma = \uparrow, \downarrow}} t_{xy} \hat{C}_{x\sigma}^\dagger \hat{C}_{y\sigma} + \underbrace{U}_{\infty} \sum_{x \in \Lambda} \hat{N}_{x\uparrow} \hat{N}_{x\downarrow} \quad (1)$$

We assume  $t_{xx} = 0$  ( $x \in \Lambda$ ) for notational simplicity

(our proof is based on Tasaki (1989))



Yosuke Nagaoka (1933-2024)  
picture from JPS website

extension is easy

## § Weak version of Nagaoka's theorem

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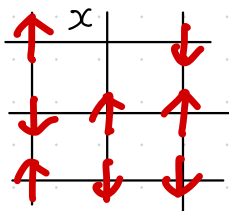
Theorem consider the Hubbard model with  $t_{xy} \geq 0$  for all  $x \neq y$ ,  $U = \infty$ , and  $N = |\Lambda| - 1$ . then there are ground states with  $S_{\text{tot}} = N/2$

- there are ferromagnetic ground states!
- but there may be ground states with smaller  $S_{\text{tot}}$

proof

$|\Phi_{x,\sigma}\rangle$  basis state where  $x$  is empty and

site  $u \in \Lambda \setminus \{x\}$  is occupied by an electron with spin  $\sigma_u = \uparrow, \downarrow$



the "hole" is at  $x \in \Lambda$

$\sigma = (\sigma_u)_{u \in \Lambda \setminus \{x\}}$  spin configuration

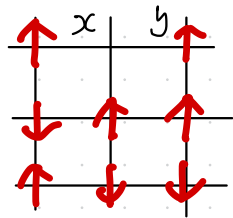
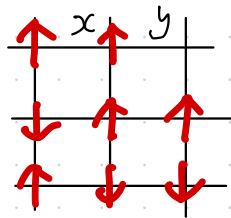
let  $\sigma'_u = \sigma''_u = \sigma_u$  for  $u \in \Lambda \setminus \{x\}$ ,  $\sigma'_x = \uparrow$ ,  $\sigma''_x = \downarrow$

$$|\Phi_{x,\sigma}\rangle = \hat{C}_{x\uparrow} \left( \prod_{u \in \Lambda} \hat{C}_{u\sigma'_u}^\dagger \right) |\Phi_{\text{vac}}\rangle = \hat{C}_{x\downarrow} \left( \prod_{u \in \Lambda} \hat{C}_{u\sigma''_u}^\dagger \right) |\Phi_{\text{vac}}\rangle \quad (1)$$

products with a fixed order

two equivalent expressions

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electron hops from  $y$  into  $x$  $|\Phi_{x,\sigma}\rangle$  $\sigma$  $|\Phi_{y,\sigma_{y\rightarrow x}}\rangle$  $\sigma_{y\rightarrow x}$ 

$$\begin{cases} (\sigma_{y\rightarrow x})_u = \sigma_u & u \neq x, y \\ (\sigma_{y\rightarrow x})_x = \sigma_y \end{cases}$$

spin configuration after the hop

when  $\sigma_y = \uparrow$ 

$$\sigma'_x = \sigma'_y = \uparrow$$

$$\sigma_y$$

$$\left( \sum_{\sigma=\uparrow,\downarrow} \hat{c}_{x\sigma}^\dagger \hat{c}_{y\sigma} \right) |\Phi_{x,\sigma}\rangle = \hat{c}_{x\uparrow}^\dagger \hat{c}_{y\uparrow} \hat{c}_{x\uparrow} \left( \prod_{u \in \mathcal{L}} \hat{c}_{u\sigma_u}^\dagger \right) |\Phi_{vac}\rangle$$

$$- \hat{c}_{y\uparrow} \hat{c}_{x\uparrow}^\dagger \hat{c}_{x\uparrow} = - \hat{c}_{y\uparrow} \hat{n}_{x\uparrow}$$

$$= - \hat{c}_{y\uparrow} \left( \prod_{u \in \mathcal{L}} \hat{c}_{u\sigma_u}^\dagger \right) |\Phi_{vac}\rangle = - |\Phi_{y,\sigma_{y\rightarrow x}}\rangle \quad (1)$$

exactly the same when  $\sigma_y = \downarrow$ 

$$\langle \Phi_{y,\tau} | \hat{H}_{hop} | \Phi_{x,\sigma} \rangle = \begin{cases} -t_{xy} & \text{if } \tau = \sigma_{y\rightarrow x} \\ 0 & \text{otherwise} \end{cases}$$

(2)

$|\Phi_{GS}\rangle$  an arbitrary normalized ground state

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$$|\Phi_{GS}\rangle = \sum_{\substack{x \in \Lambda \\ \sigma \in \mathcal{S}_x}} \varphi_{x,\sigma} |\Phi_{x,\sigma}\rangle \quad (1) \quad \mathcal{S}_x: \text{spin configurations on } \Lambda \setminus \{x\}$$

we can assume  $\varphi_{x,\sigma} \in \mathbb{R}$

$$\text{let } \Xi_x = \sqrt{\sum_{\sigma \in \mathcal{S}_x} (\varphi_{x,\sigma})^2} \quad (2) \quad \text{and } |\Phi_{\uparrow}\rangle = \sum_{x \in \Lambda} \Xi_x |\Phi_{x,(\uparrow)}\rangle \quad (3)$$

has  $S_{\text{tot}} = \frac{N}{2}$

all up

$$E_{GS} = \langle \Phi_{GS} | \hat{H}_{\text{hop}} | \Phi_{GS} \rangle = \sum_{\substack{x,y \in \Lambda \\ (x \neq y)}} \sum_{\substack{\sigma \in \mathcal{S}_x \\ \tau \in \mathcal{S}_y}} \varphi_{y,\tau} \varphi_{x,\sigma} \langle \Phi_{y,\tau} | \hat{H}_{\text{hop}} | \Phi_{x,\sigma} \rangle$$

$$\langle \Phi_{\uparrow} | \Phi_{\uparrow} \rangle = 1$$

$$\stackrel{\text{p3-(2)}}{=} - \sum_{\substack{x,y \in \Lambda \\ (x \neq y)}} t_{xy} \sum_{\sigma \in \mathcal{S}_x} \varphi_{y,\sigma_{y \rightarrow x}} \varphi_{x,\sigma} \geq - \sum_{\substack{x,y \in \Lambda \\ (x \neq y)}} t_{xy} \sqrt{\sum_{\sigma \in \mathcal{S}_x} (\varphi_{y,\sigma_{y \rightarrow x}})^2 \sum_{\sigma \in \mathcal{S}_x} (\varphi_{x,\sigma})^2}$$

$$= - \sum_{\substack{x,y \in \Lambda \\ (x \neq y)}} t_{xy} \Xi_y \Xi_x = \langle \Phi_{\uparrow} | \hat{H}_{\text{hop}} | \Phi_{\uparrow} \rangle$$

$|\Phi_{\uparrow}\rangle$  is also a ground state! //

## § connectivity condition and Nagaoka's theorem

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a Hubbard model with  $U=\infty$  and  $N=|\Lambda|-1$  is said to satisfy the connectivity condition if all basis states  $|\Phi_{x,\sigma}\rangle$  with common  $S_{\text{tot}}^{(z)} = \sum_{u \in \Lambda \setminus \{x\}} \sigma_u$  are "connected" via nonzero matrix elements of  $\hat{H}_{\text{hop}}$  → P3-(2)

Theorem consider the Hubbard model with  $t_{xy} \geq 0$  for all  $x \neq y$ ,  $U=\infty$ , and  $N=|\Lambda|-1$ , where the connectivity condition is satisfied. then the model exhibits ferromagnetism and the ground state is unique

proof Tasaki 1989

part 3 - p.11

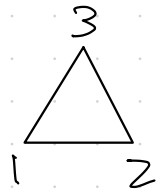
the matrix with entries  $\langle \Phi_{y,\tau} | \hat{H}_{\text{hop}} | \Phi_{x,\sigma} \rangle$  (within a fixed  $S_{\text{tot}}^{(z)}$  sector)

satisfies the conditions for the Perron-Frobenius theorem (appendix 1)

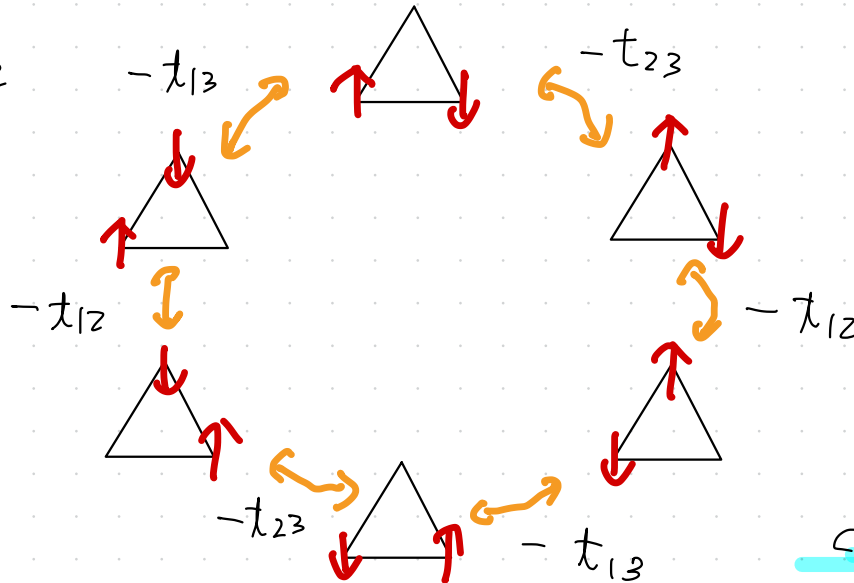
thus the ground state is unique, but it must have  $S_{\text{tot}} = \frac{N}{2}$  from the weak version //

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a Hubbard model with  $U=\infty$  and  $N=|\Lambda|-1$  is said to satisfy the connectivity condition if all basis states  $|\Phi_{x,\sigma}\rangle$  with common  $S_{\text{tot}}^{(z)} = \sum_{u \in \Lambda \setminus \{x\}} \sigma_u$  are "connected" via nonzero matrix elements of  $\hat{H}_{\text{hop}}$



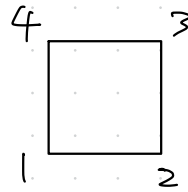
$$\Lambda = \{1, 2, 3\}, \quad N = 2, \quad S_{\text{tot}}^{(z)} = 0$$



you can reach any configuration by moving the "hole"

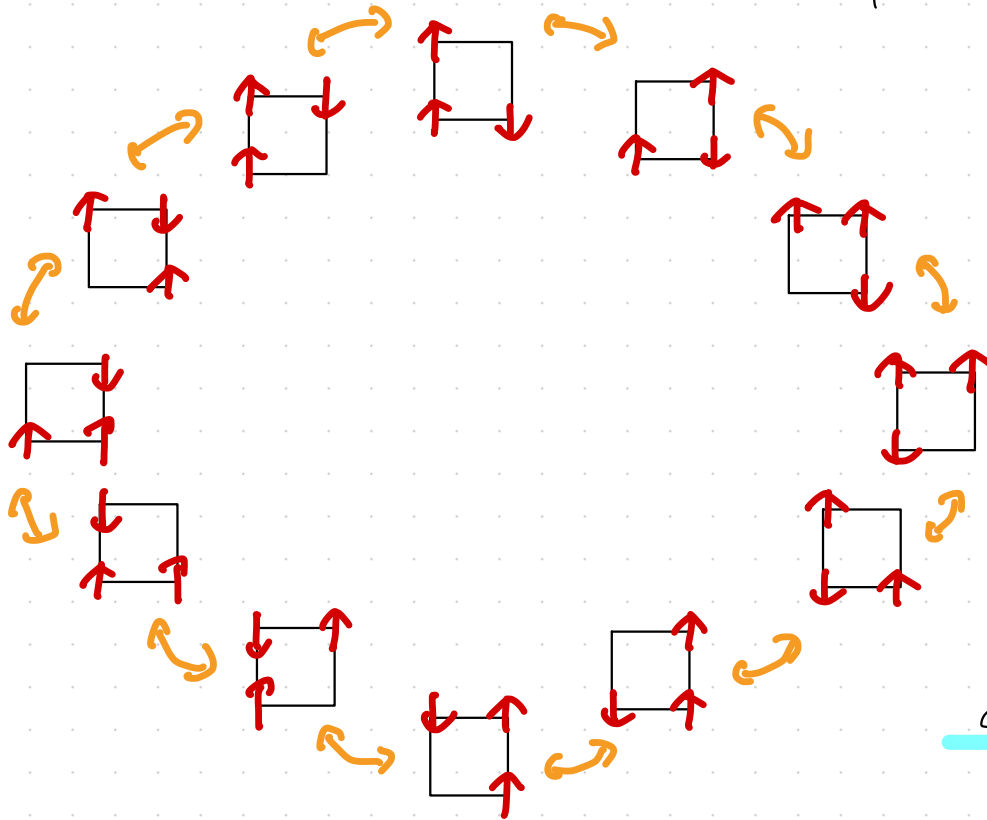
connectivity condition holds!

$$\Lambda = \{1, 2, 3, 4\}, N = 3, \hat{S}_{\text{tot}}^{(z)} = \frac{1}{2}$$



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$$t_{12}, t_{23}, t_{34}, t_{41} > 0$$



you can reach any configuration by moving the "hole"

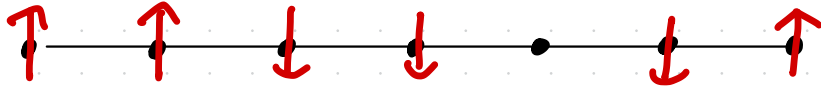
connectivity condition holds!



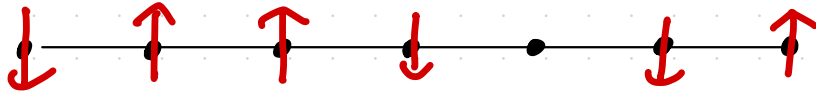
connectivity condition may fail

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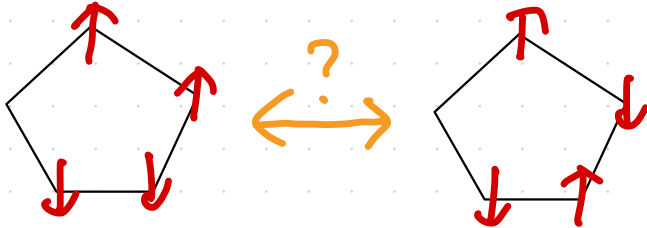
1D chain with nearest neighbor hopping and open boundaries



↓ never be connected to



how about



# proof of connectivity condition for the square lattice

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one can move the "hole" to any place

→ nearest neighbor hopping

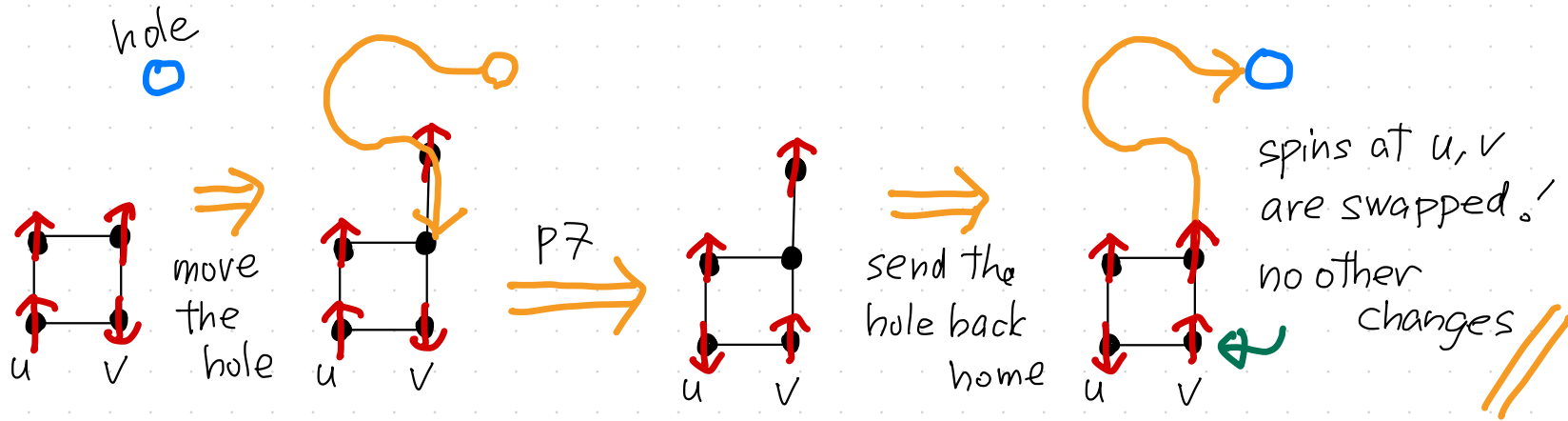
it suffices to show  $|\Phi_{x,\emptyset}\rangle$  is connected to  $|\Phi_{x,\pi}\rangle$

if  $\sum \sigma_u = \sum \tau_u \rightarrow \emptyset$  is a permutation of  $\pi$

We only need to show  $|\Phi_{x,\emptyset}\rangle$  is connected to  $|\Phi_{x,\sigma_{u \leftrightarrow v}}\rangle$

for any neighboring  $u, v$ .

↙ swap spins at  $u$  and  $v$



## § Nagaoka's ferromagnetism

► discovered in 1966, and had remained the only rigorous example of ferromagnetism in the Hubbard model for more than two decades

► theoretically interesting, but, physically, a bit too artificial  
the motion of a single "hole" stabilizes the order of the whole system

<sup>7</sup>In Japanese institutions it often happens that the presence of a single foreign participant in a seminar room makes everybody shift from Japanese to English. I used to mention this phenomenon in the introduction to talks about Nagaoka's ferromagnetism. I remember one seminar at RIMS in Kyoto, where I had a perfect situation for this joke; the topic was about my refinement of Nagaoka's theorem, there was exactly one foreign participant at the seminar, and among many Japanese participants was Yosuke Nagaoka!

Hal Tasaki "Physics and Mathematics of Quantum Many-Body Systems"

► likely to be unstable against addition of holes or finite  $U > 0$